Computer Science & IT

Data Structure & Programming

Tree

Lecture No. 06

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Recap of Previous Lecture

Topic BST Insert

Topic Delete, search

Topic Min Max

Topic

Topic

Slide

Topics to be Covered

- **Topic** AVL tree
- **Topic** Balancing factor
- **Topic** Rotation
- **Topic**
- **Topic**

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Topic : Tree

AVL tree is Height balanced BST

Balancing factor of node is 0, 1 or -1

Balancing factor : Height of left subtree - Height of Right subtree

or

Height of right subtree - left subtree

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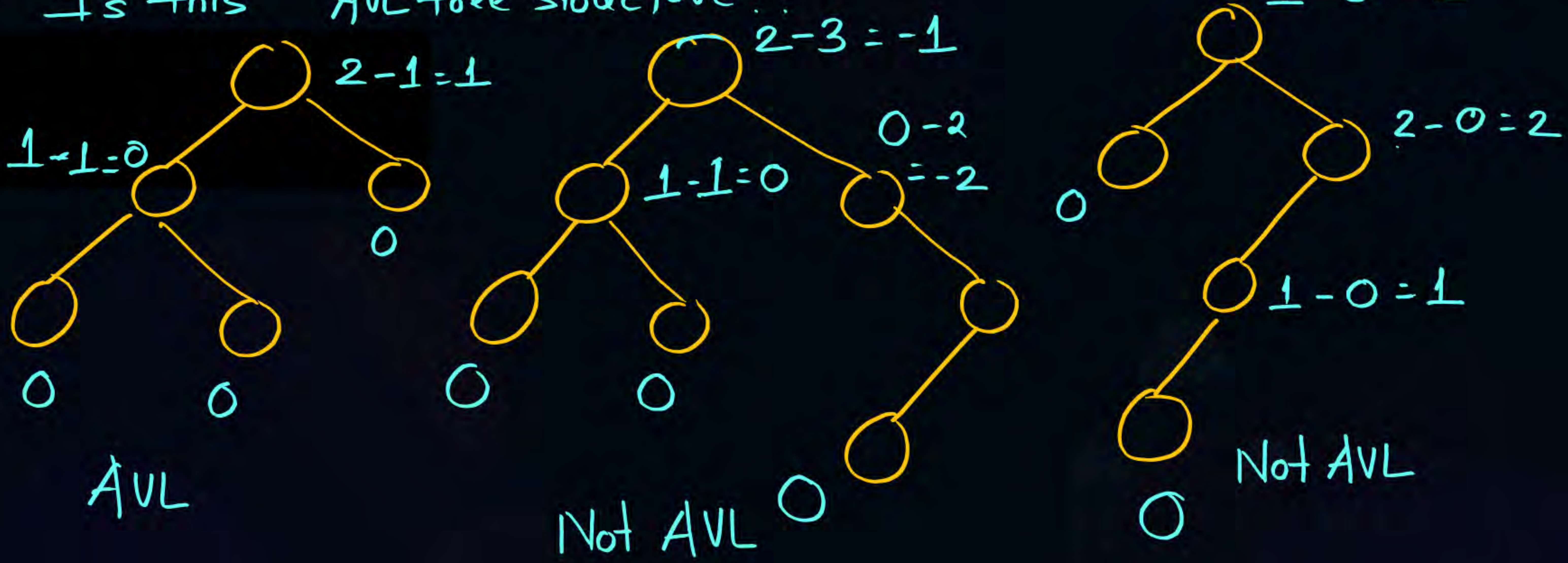
Topic : Tree

Is this AVL tree structure??



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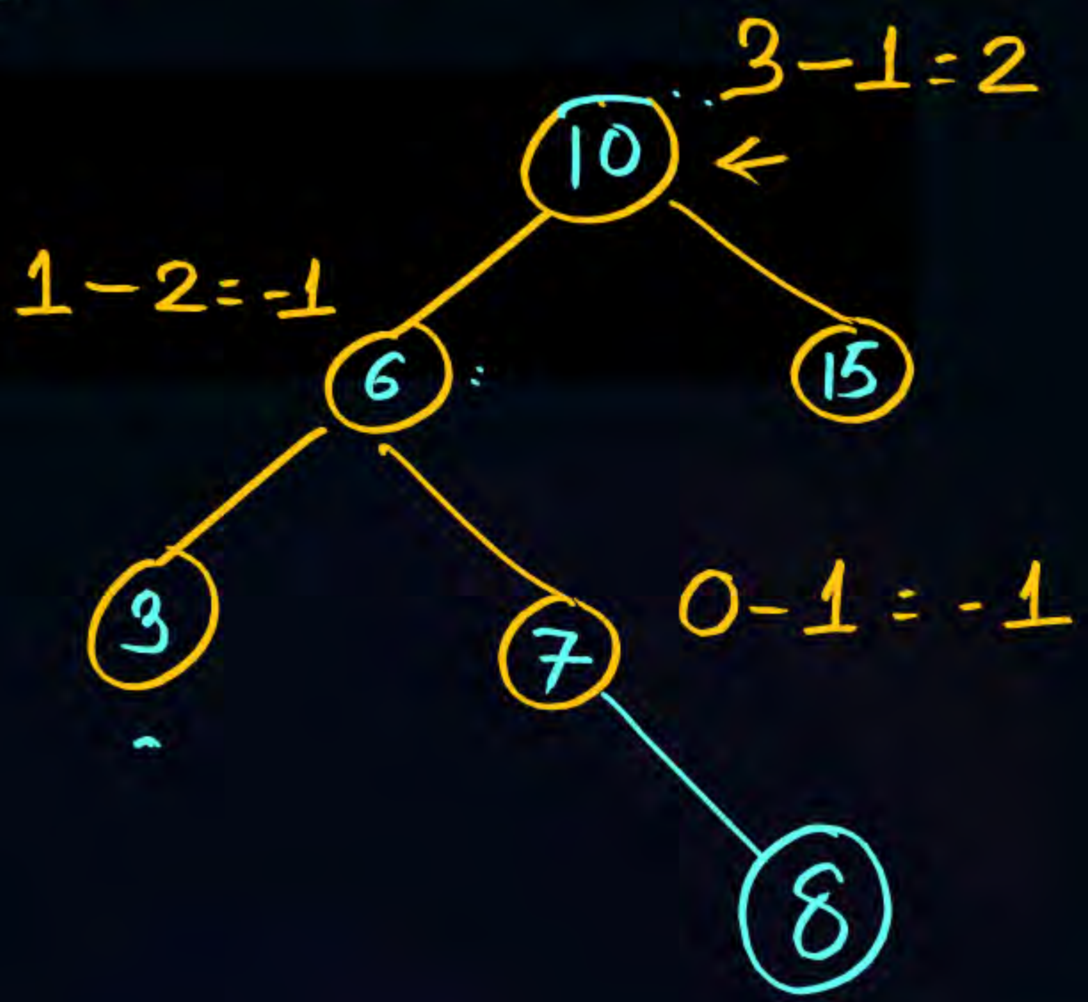
Topic : Tree



Iseot 8 if balonce

we need to find cortical Node

Cortical Node : In Newly inserted node to root path the first node dissatisfy balancing property is called Cortical Node.

10 Cortical Node

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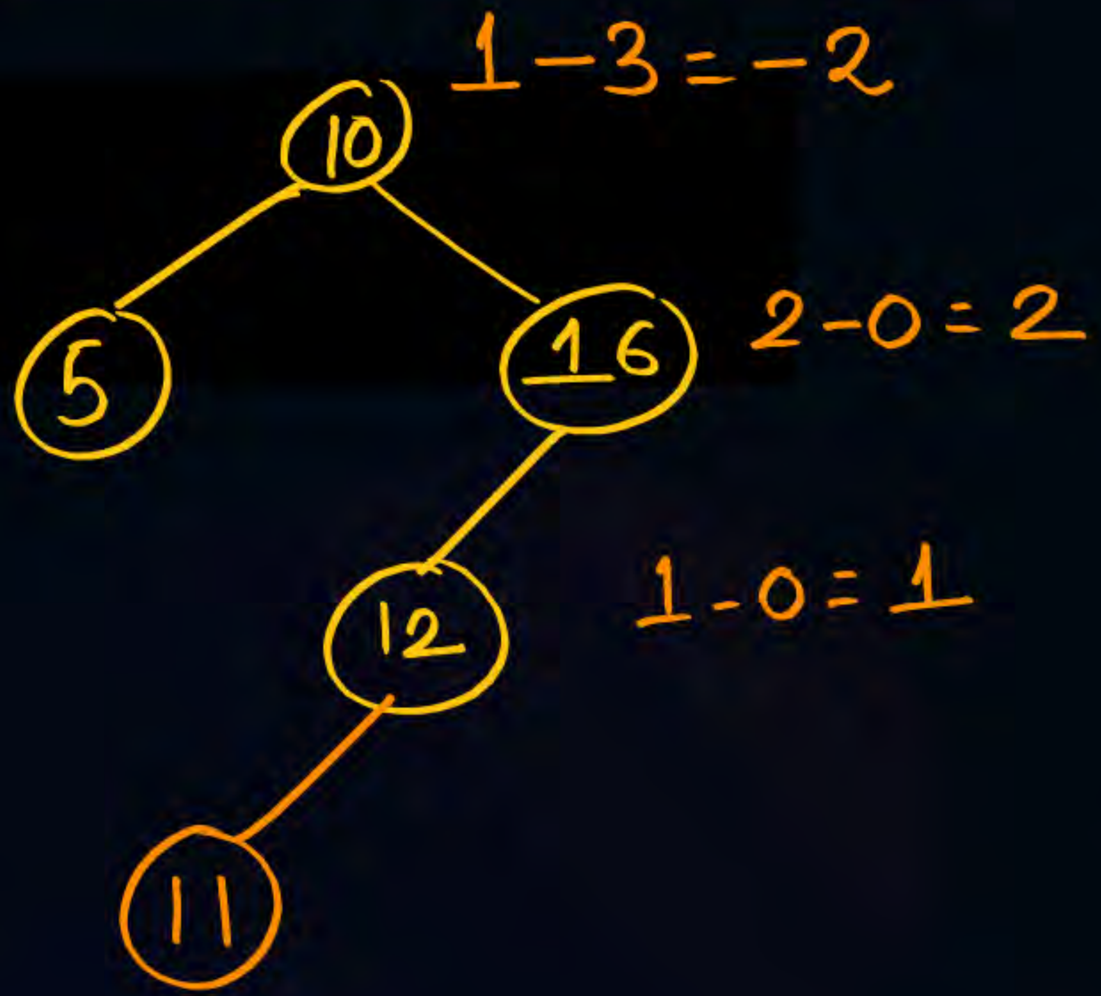
Topic : Tree



Insert 11 in the tree

10 and 16 both violates balancing property

first node in newly inserted node to root path

16 is critical Node.

To balance it we need to perform Rotation

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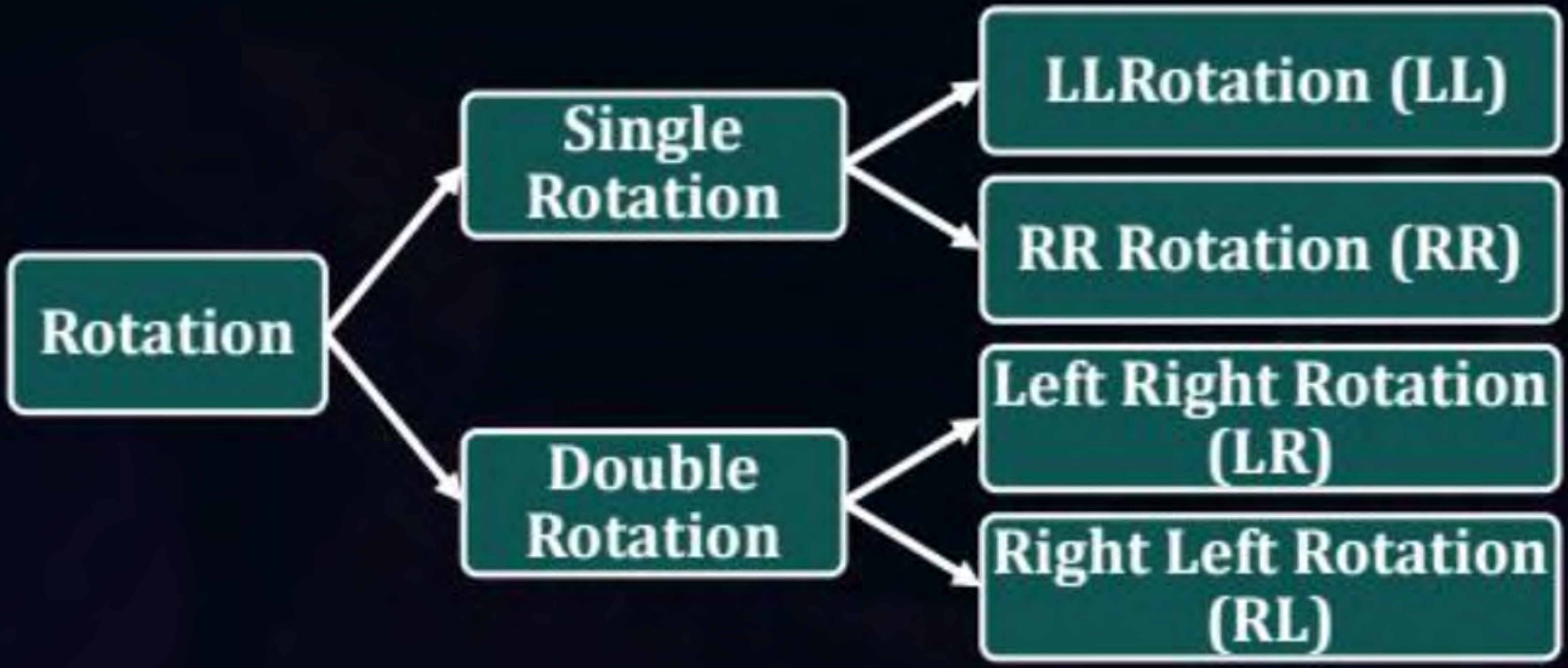
Topic : AVL Tree

To balance itself, an AVL tree may perform the following four kinds of rotations



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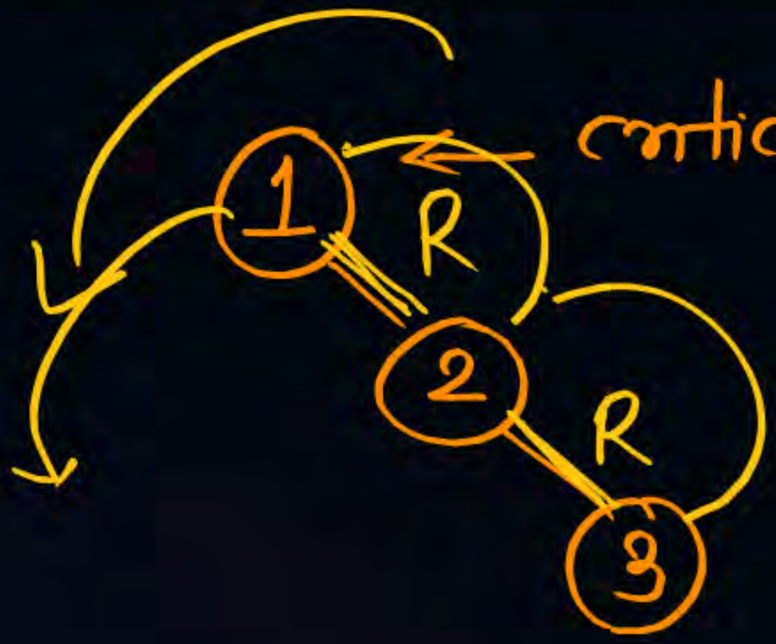
Topic : RR Rotation

Insert 1,2,3

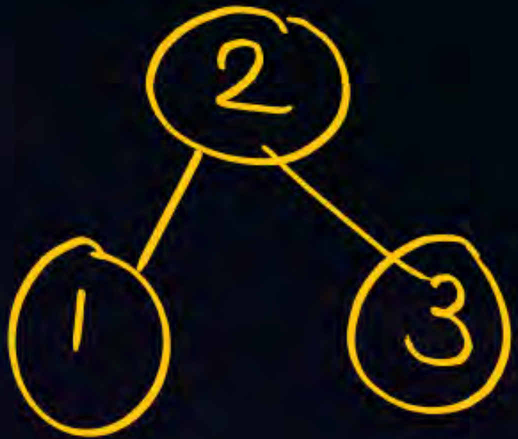


critical Node 1

from critical Node to newly inserted draw 2 edges

(2 edge may or may not reach to new node)

1. Left Rotation will fix



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Topic : RR Rotation

Insert 1,2,3

Slide

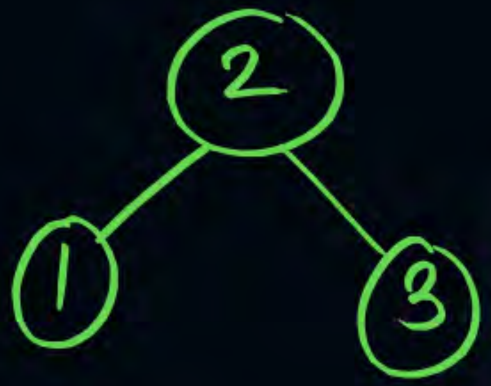
Topic : AVL Tree

In LL Rotation

Insert 3,2,1

← critical Node

Right Rotation will fix



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Topic : AVL Tree

In LL Rotation Insert 3,2,1

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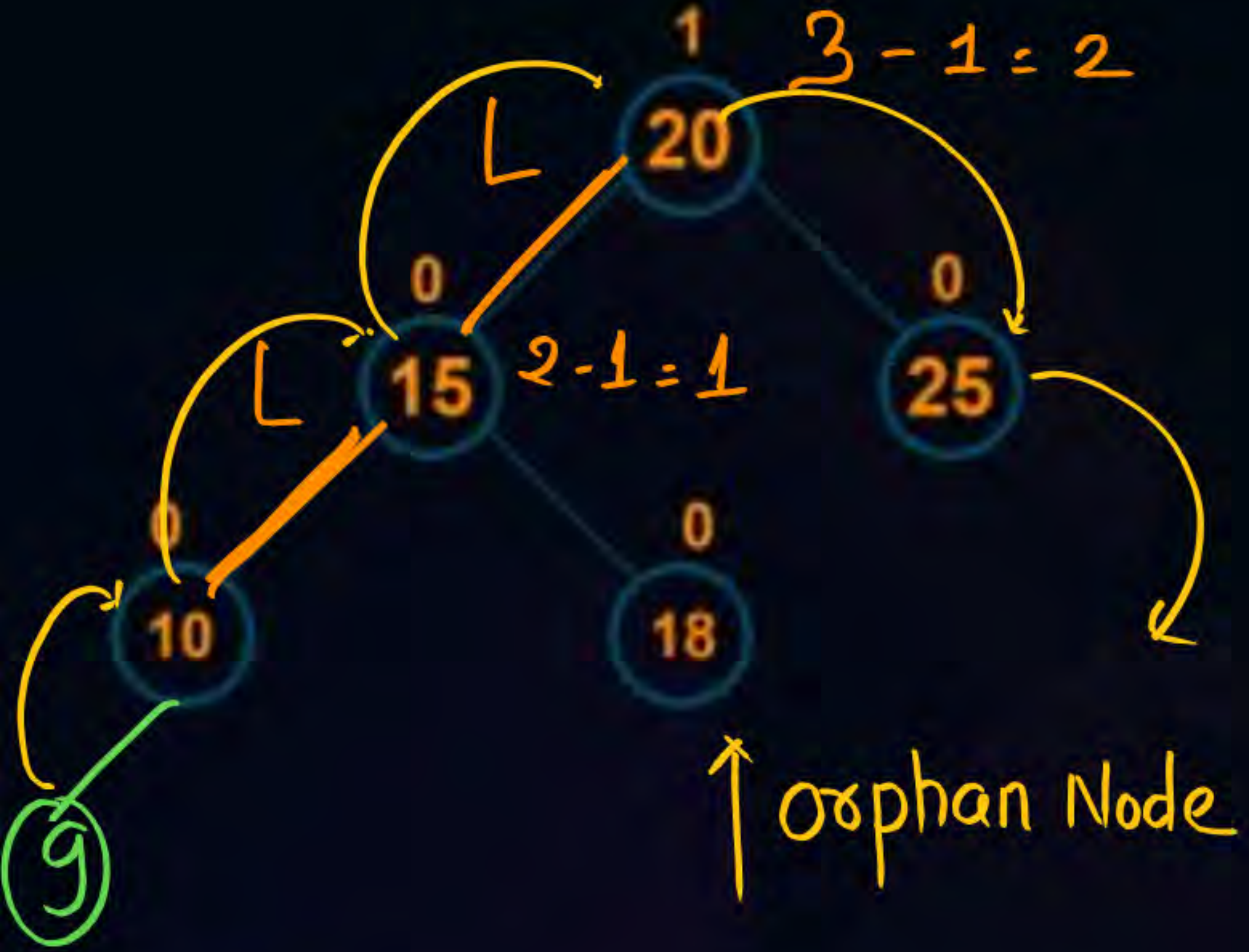
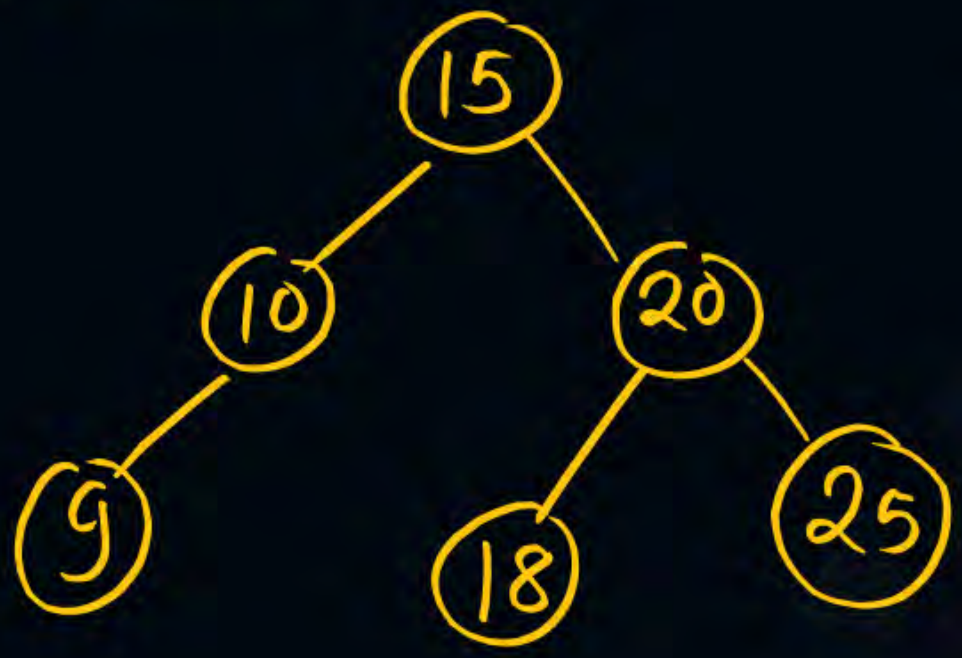
Topic : AVL tree

More Situation

Insert 9

20 critical Node





Orphan node

Rotating Node is 20

orphan Node is in leftsubtree of rotating Node after rotation 18 will be leftchild of 20

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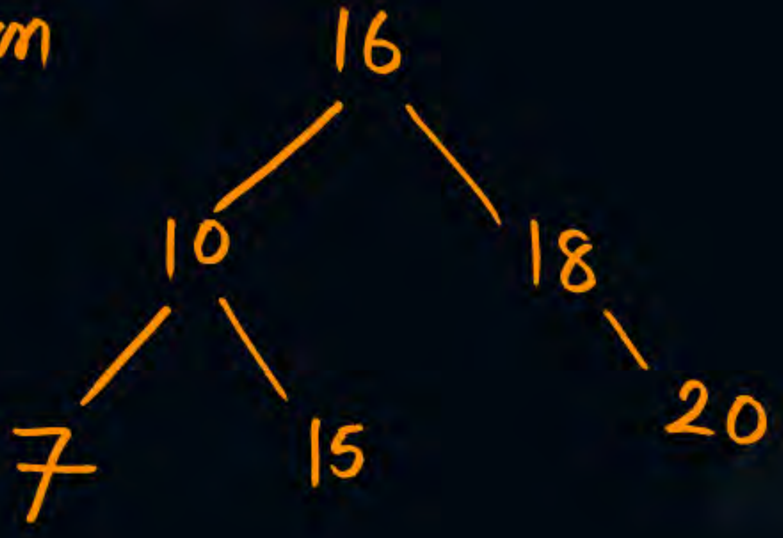
Topic : LR Rotation

Left Rotation



15 is in right sub tree of 10 after rotation it will become right child of 10 (Rotating Node)

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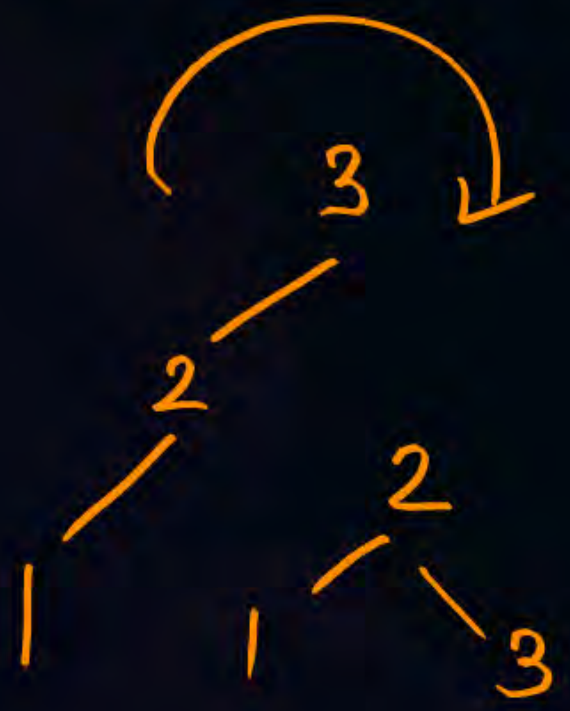
Topic : LR Rotation

Insert ~~3,2,1~~ 3,1,2

← critical Node L R



Memorization :

L : Left child of critical Node Rotated Left (RR Rotation)

R : Critical Node will be Rotated right (LL Rotation)

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Topic : LR Rotation

Insert 3,2,1

Slide

Topic : The RL Rotation

Insert 1,3,2

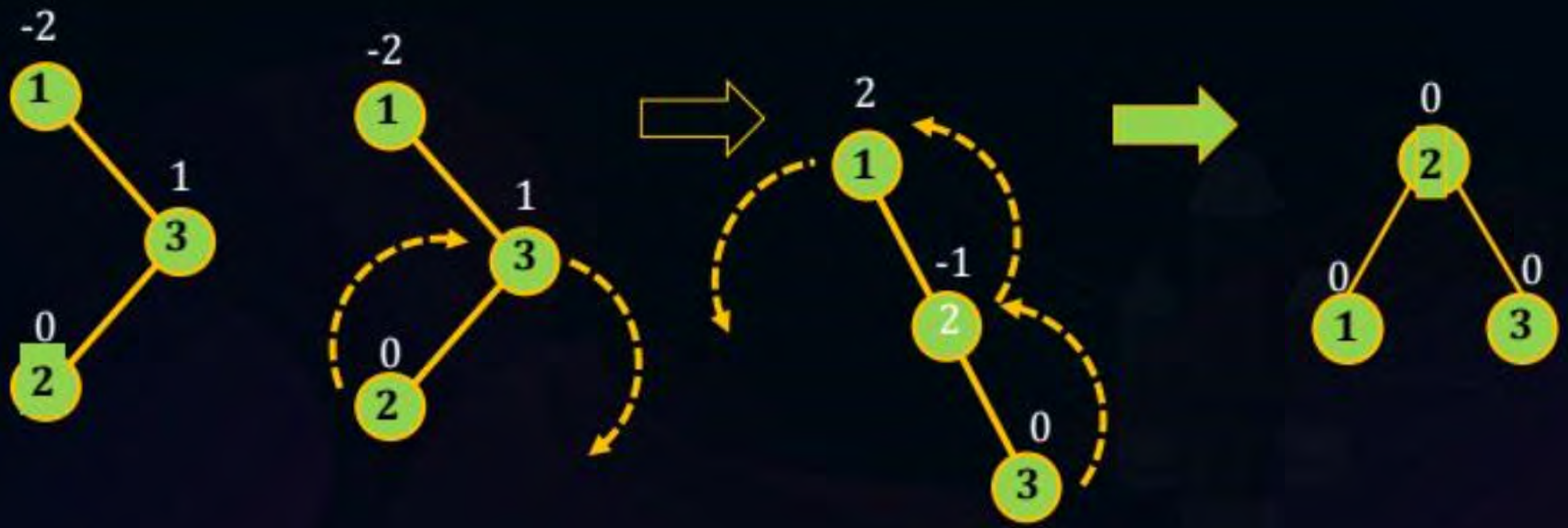
R : Right child of critical Node Rotated right (LL)

L. Critical Node will be rotated left. (RR)

Slide

Topic : The RL Rotation

Insert 1,3,2



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Topic : AVL Tree

Insert the following sequence of element in AVL tree

3, 2, 1, 4, 5, 6, 7, 15, 16, 14, 13

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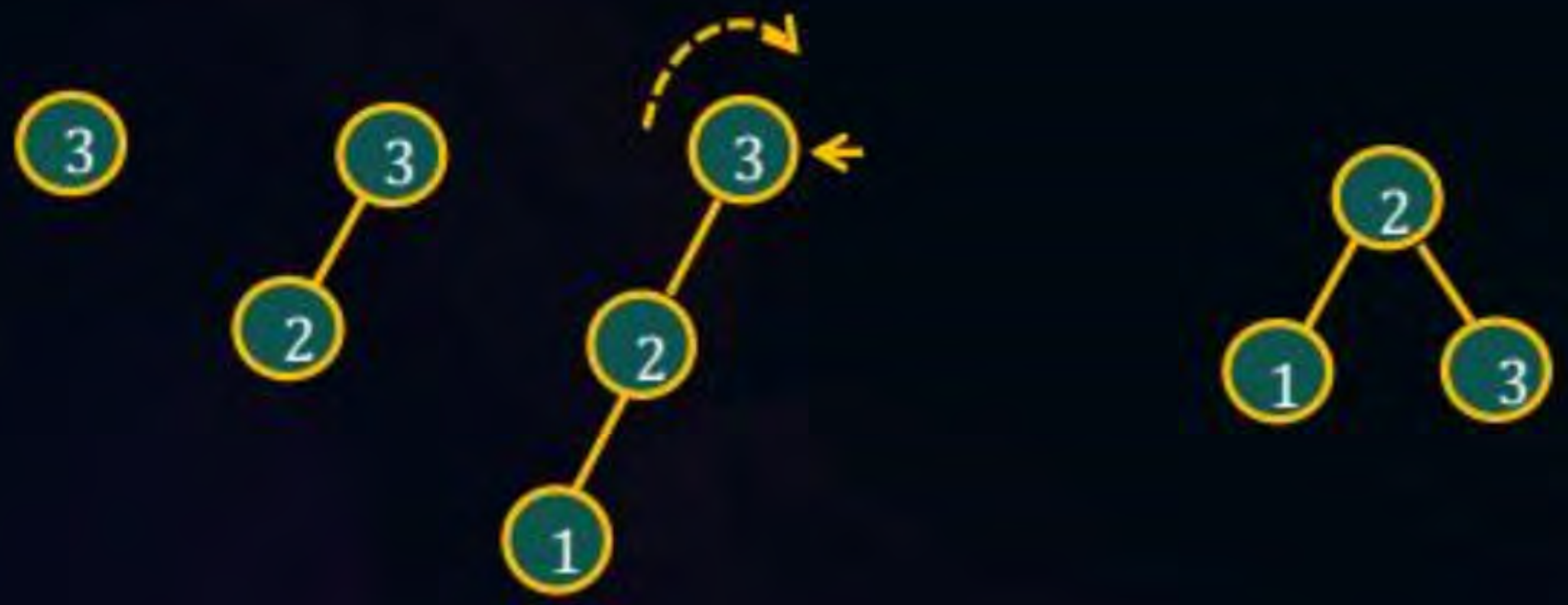
Topic : AVL Tree

Insert the following sequence of element in AVL tree
3, 2, 1, 4, 5, 6, 7, 15, 16, 14, 13



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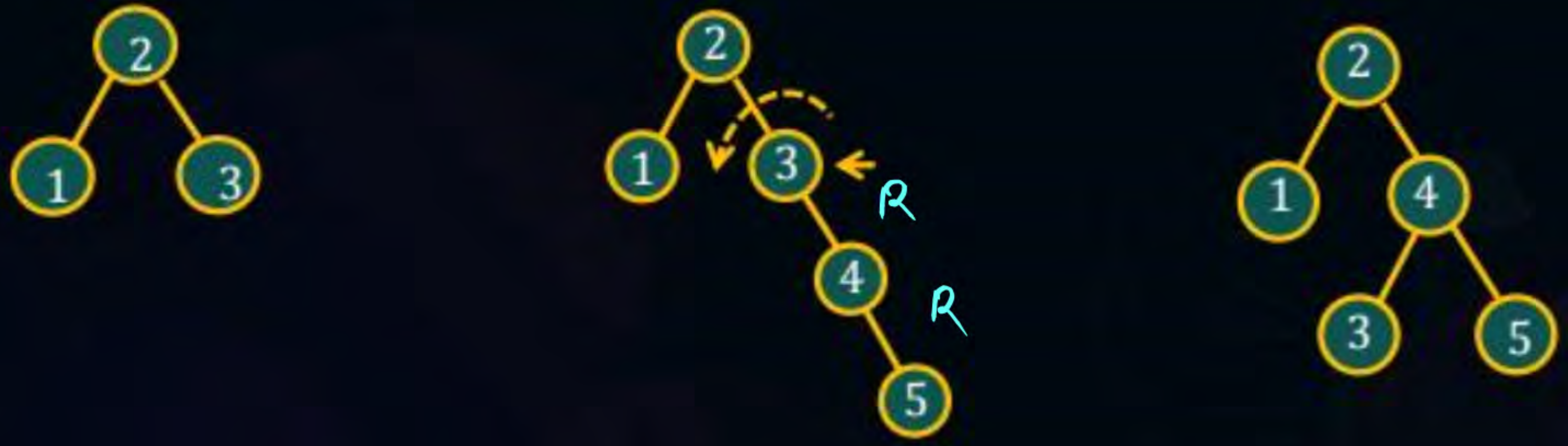
Topic : AVL Tree

Insert 4, 5



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Insert 6

Topic : AVL Tree

Insert 7

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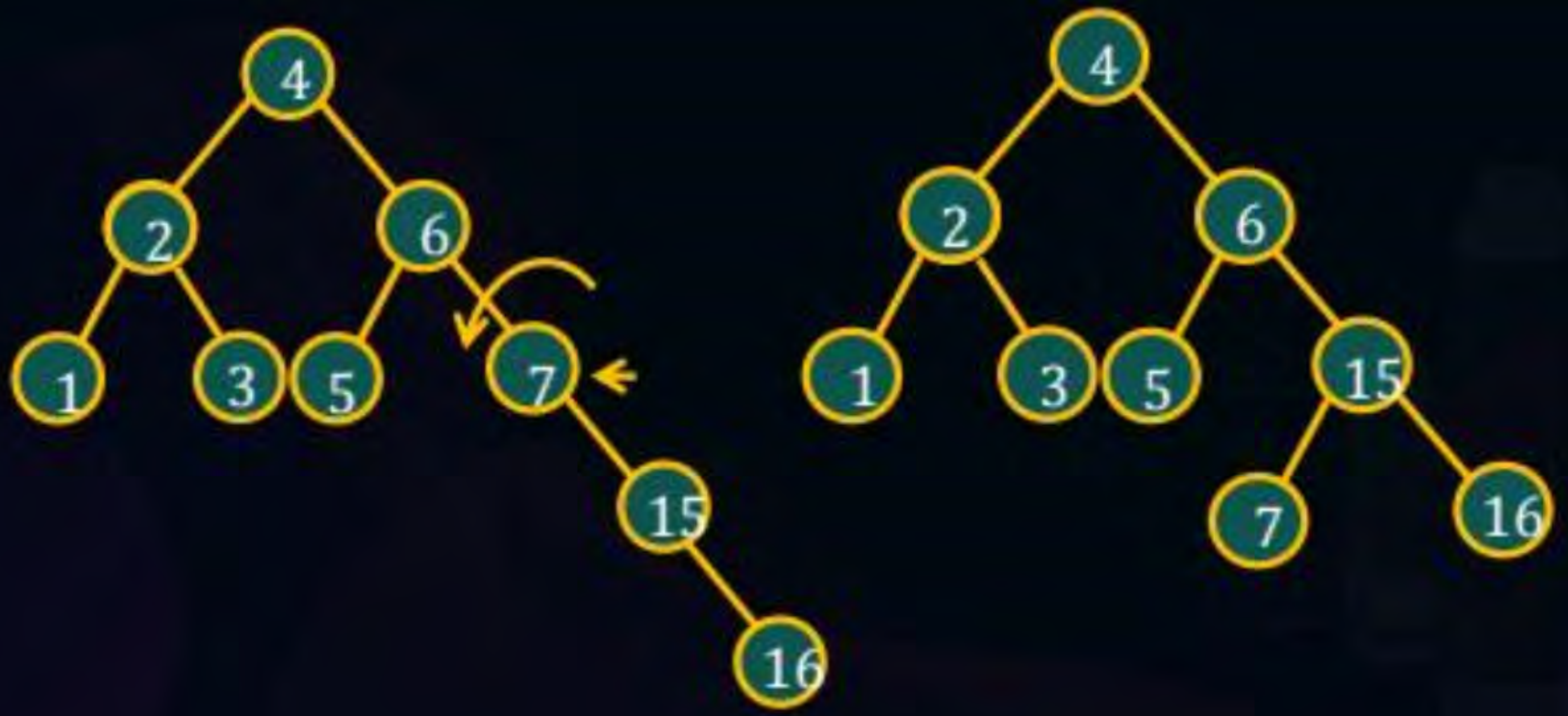
Topic : AVL Tree

Insert 15, 16



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Topic : AVL Tree

Insert 14

Slide

Insert 13

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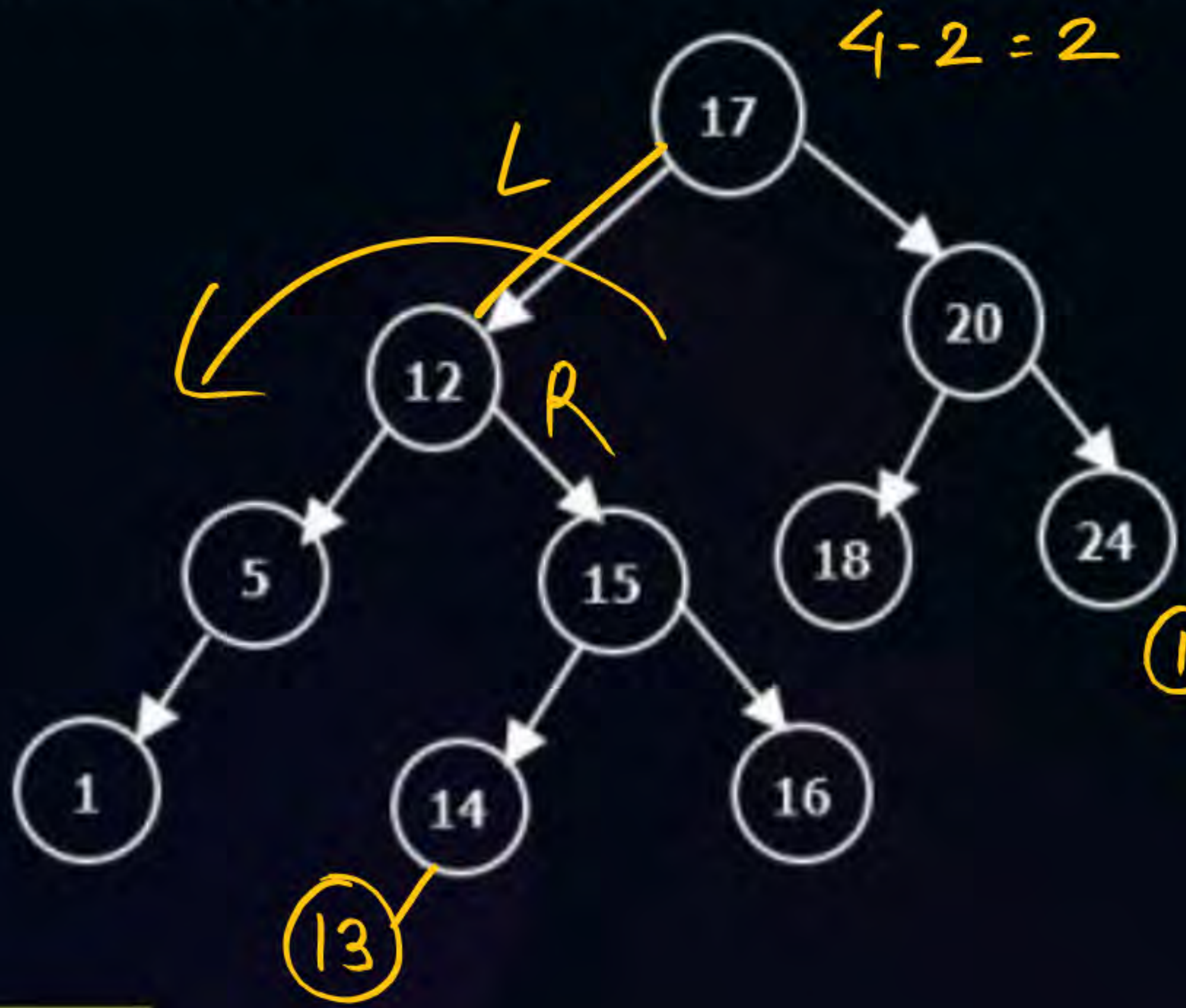
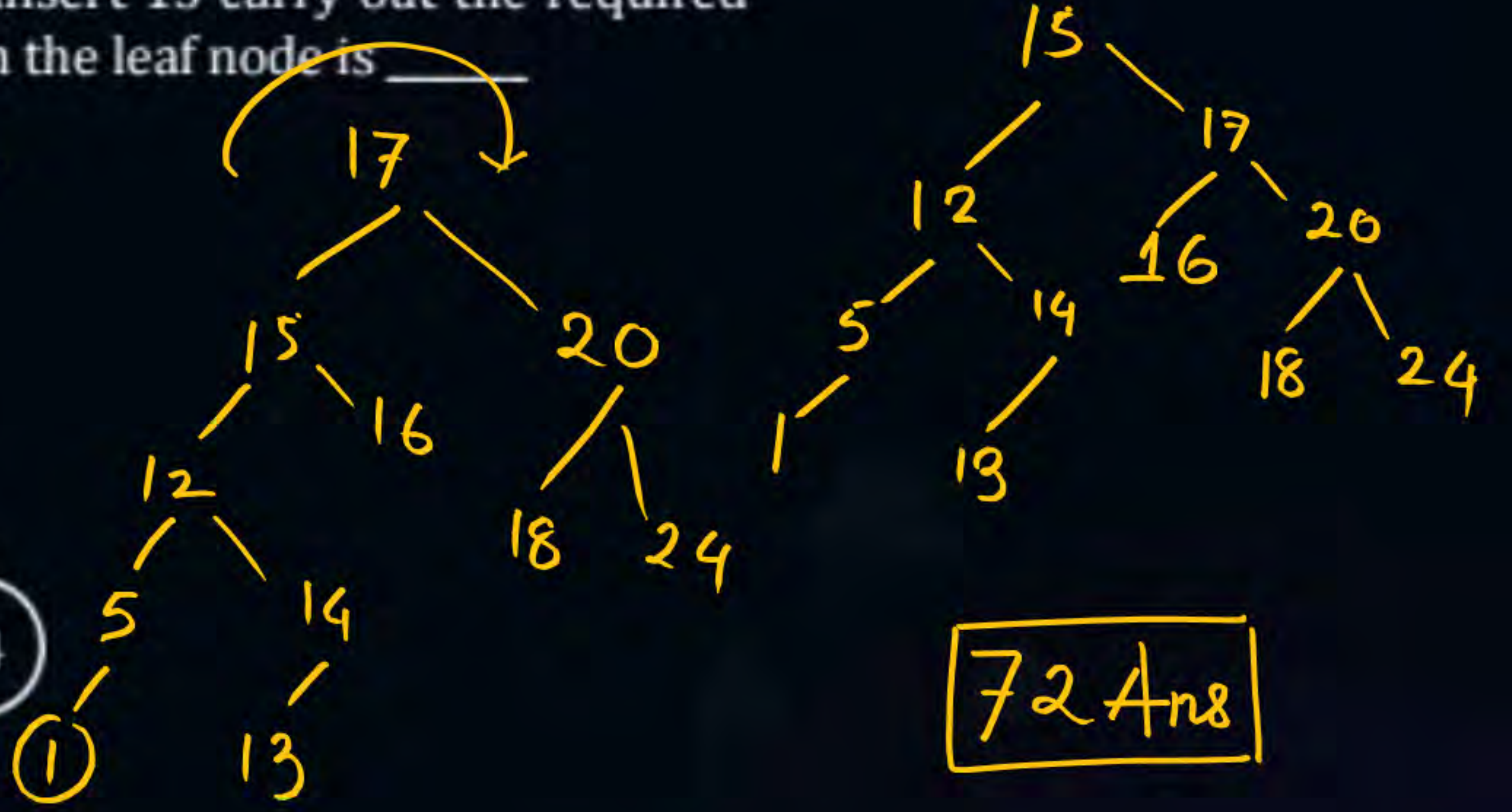
Topic : AVL Tree

#Q Consider the tree given in the below figure, insert 13 carry out the required rebalancing of the tree and find sum of element in the leaf node is ______





$\boxed{72 \text{ Ans}}$

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Question

10

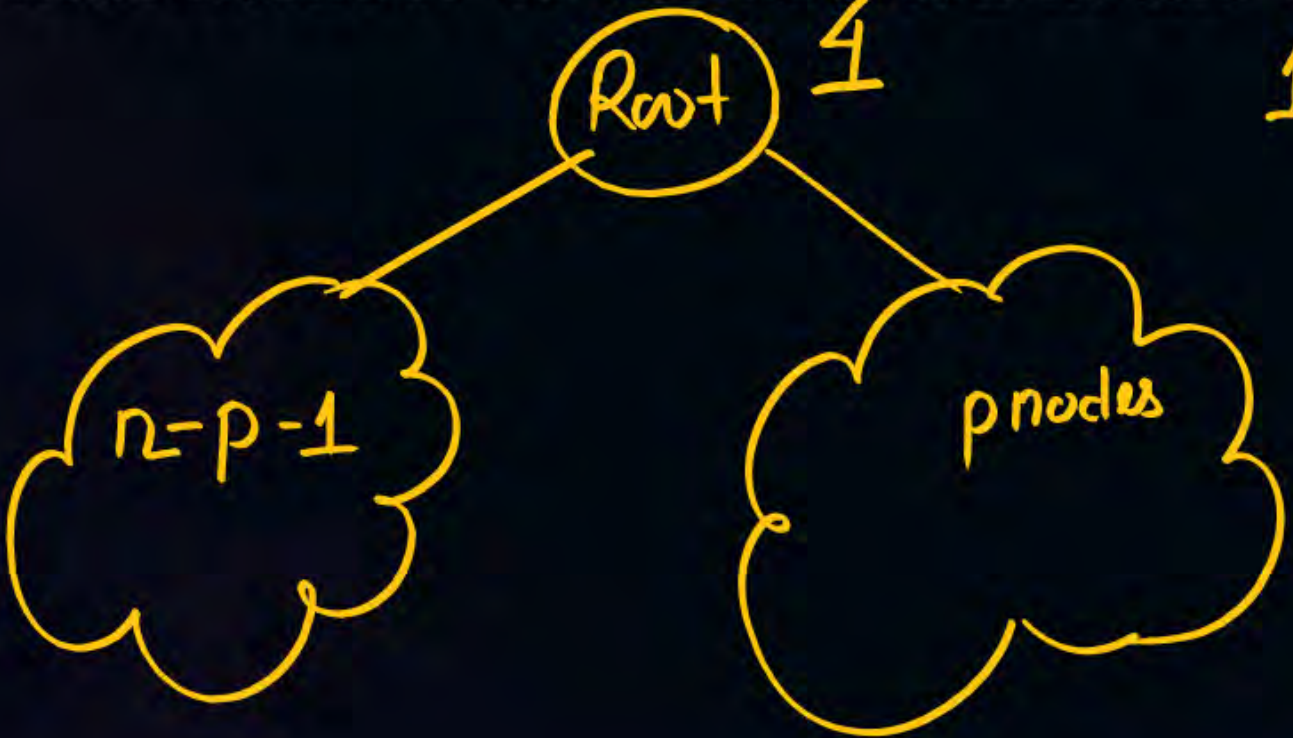
#Q The numbers 1, 2, ...n are inserted in a binary search tree in some order. In the resulting tree, the right subtree of the root contains p nodes. The first number to be inserted in the tree must be.

(A) p

(B) p + 1

(C) n - p ✓

(D) n - p + 1



1 2 ... p ... n

n-p

n=10

p=6

5,6,7,8,9,10

n-p n-p+1 ... n

$\cancel{n}-\cancel{p}+p-\cancel{1}$

Question

Consider the following intermediate representation of binary search tree in which all keys are distinct and m and n are key values.

$m - 1$ to $n + 1$

$m - 1 - n - \cancel{1} + \cancel{1}$

$\boxed{m - n - 1}$

The maximum number keys that can be inserted on right hand side of n?

(A) $m + n$ (B) $m+n-1$ (C) $m-n-1$ (D) $m-n+1$

5

7, 8, 9

Topic : Question

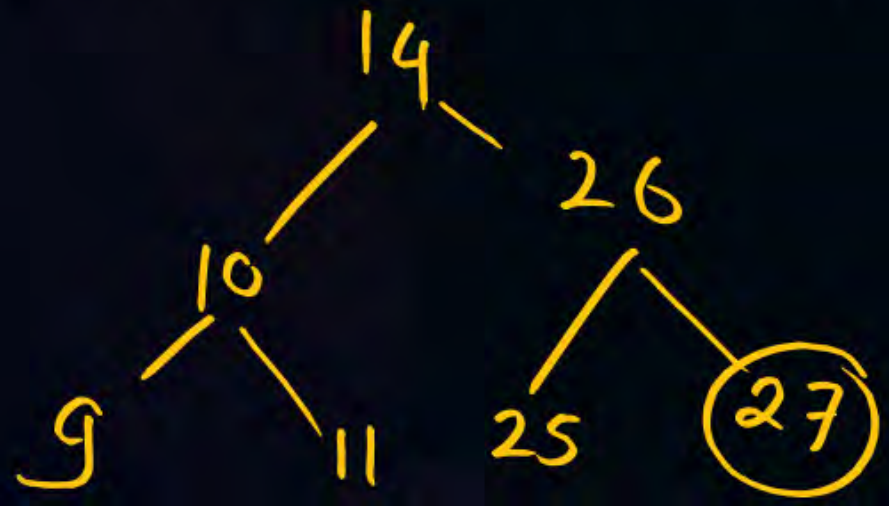
Q. A binary search tree T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T?

(A) $\Theta(n)$

(B) $\Theta(n\log n)$

(C) $\Theta(\log n)$

(D) $\Theta(1)$



(I) if t→righ != NULL
return t→data
else
if (t→left != NULL)
return t→left→data
els return -1

THANK - YOU

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